Future Me: Time-Locked Messaging System

AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION

MINI PROJECT REPORT

Submitted by

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CHENNAI - 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled "Future Me: Time-Locked Messaging System " is the Bonafide work of ARUL SANTHOSH X (241501022) and ANBARASU V (241501018) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted to Project Viva-Voce Examination held on …………………

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training, and research.

DEPARTMENT MISSION

- To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- To promote research activities amongst the students and the members of faculty that could benefit the society.
- To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To equip students with essential background in computer science with emphasis on Artificial Intelligence, Machine Learning, basic electronics, and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications using emerging technologies.

PEO 3: To encourage research and innovative project development in the field of Artificial Intelligence, Machine Learning, Deep Learning, Networking, Security, Web development, Data Science, and emerging technologies for social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate:

PSO 1: Foundation Skills: Ability to understand, analyse and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational tasks, model real world problems using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically socially responsible AI and ML professional.

COURSE OBJECTIVES

To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.

- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server-Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOMES

On completion of course students will be able to

CO 1: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.

CO 2: Design and implement dynamic web page with validation using JavaScript objects and by applying different event handling mechanism.

CO 3: Design and implement simple webpage to learn JSP and Servlet.

CO 4: Design and implement simple Application Design.

CO 5: Identify various concepts of mobile programming that make it unique from programming for other platforms

CO - PO — PSO matrices of course

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
A123431.01	3	3	3	3	3	3	2	2	3	2	1	3	3	3	2
A123431.02	3	3	3	3	3	3	2	2	3	2	1	1	3	3	2
A123431.03	3	3	3	3	3	3	2	2	3	2	2	2	3	3	3
A123431.04	3	3	3	3	3	3	2	2	3	2	2	3	3	3	3
A123431.05	3	3	3	3	3	3	2	2	3	2	3	3	3	3	3
Average	3	3	3	3	3	3	2	2	3	2	1.8	2.1	3	3	2.4

Note: Enter correlation levels l, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “- “

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	**Quality Education**	The course provides fundamental knowledge in web and mobile application development, enabling students to gain practical technical skills, improve problem-solving abilities, and become industry-ready professionals.
SDG-8	**Decent Work and Economic Growth**	The subject equips students with job-oriented skills such as web development, client-side and server-side scripting, and mobile application design, increasing employability and supporting economic growth.
SDG-9	**Industry, Innovation, and Infrastructure**	The course promotes innovation by teaching modern technologies used in developing web and mobile applications, contributing to digital infrastructure and technological advancement.
SDG-10	**Reduced Inequalities**	By enabling the development of accessible web and mobile applications, the course helps bridge the digital divide and ensures equal access to digital services for diverse users.

TABLE OF CONTENTS

ABBREVIATION	FULL FORM
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol
DOM	Document Object Model
API	Application Programming Interface
JSON	JavaScript Object Notation
URL	Uniform Resource Locator
DB	Database
CRUD	Create, Read, Update, Delete
IDE	Integrated Development Environment
VS Code	Visual Studio Code
RAM	Random Access Memory
CPU	Central Processing Unit
OTP	One-Time Password *(future enhancement)*
Auth	Authentication
LS	LocalStorage
SPA	Single Page Application

1 INTRODUCTION

1.1 General

In today's digital era, web applications play a significant role in storing and managing personal information. With advancements in frontend technologies, it has become possible to build powerful applications without relying heavily on backend systems.

This project introduces a unique concept where users can send messages to their future selves. It combines creativity with technology by allowing users to store messages that remain hidden until a specific time, thereby creating a sense of anticipation and emotional connection.

1.2 Need for Future Me Web Application

In daily life, people often forget their goals, emotions, and important thoughts due to busy schedules and changing priorities. There is a growing need for applications that help individuals reflect on their past and reconnect with their earlier thoughts.

This system fulfills that need by enabling users to store messages securely and revisit them later. It can be used for motivation, self-reflection, reminders, or even emotional journaling.

1.3 Scope of the System

The scope of this system is limited to a client-side web application that allows users to log in, create messages, set future dates, and retrieve those messages when the time arrives.

The system does not require a backend server or database, making it simple and efficient. It is designed for personal use and can run on any modern web browser.

1.4 Organization of the Report

This report is structured into multiple chapters. Chapter 2 discusses the literature survey. Chapter 3 explains the problem and objectives. Chapter 4 describes system design.

Chapter 5 covers implementation details. Chapter 6 presents results and testing. Chapter 7 concludes the report and suggests future enhancements.

🟦 2 LITERATURE SURVEY

2.1 Introduction

The literature survey focuses on analyzing existing systems related to reminders, note-taking, and delayed messaging applications. These systems provide insights into how user data is managed and how time-based features are implemented.

2.2 Existing Reminder and Messaging Systems

Applications such as Google Calendar, Notes apps, and reminder tools allow users to schedule events and store notes. However, they do not provide a dedicated feature for sending messages to the future in an engaging way.

Some email services offer delayed delivery, but they require backend support and internet connectivity, making them less efficient for lightweight applications.

2.3 Related Work

Previous studies highlight the use of LocalStorage as an effective method for storing data on the client side. JavaScript is widely used for implementing dynamic features such as countdown timers and event handling.

Authentication systems in basic web applications often rely on simple validation methods, which are sufficient for small-scale projects.

2.4 Summary

The literature survey reveals that while existing systems provide partial solutions, there is no simple and lightweight platform that combines login authentication with future message storage. This project aims to bridge that gap.

🟦 3 PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

There is a lack of simple web applications that allow users to securely store personal messages and retrieve them at a future time without requiring complex backend systems.

Most existing applications either focus on reminders or note storage but do not combine both functionalities in an engaging way.

3.2 Objectives of the System

- **To design a secure login system for user authentication**
- **To enable users to create and store messages**
- **To implement a real-time countdown timer**
- **To ensure messages remain locked until the selected date**
- **To provide an intuitive and attractive user interface**

4 SYSTEM DESIGN (DETAILED)

4.1 Introduction

System design plays a crucial role in determining how efficiently the application performs and how easily users can interact with it. The design of the *Future Me Website* focuses on simplicity, usability, and performance. Since the application is fully client-side, the design ensures that all operations are handled within the browser without requiring server communication.

The main objective of the design is to provide a smooth user experience while maintaining data persistence and logical accuracy. The system is modular, meaning each component performs a specific task, making it easier to maintain and upgrade in the future.

4.2 System Architecture

The system follows a three-layer architecture, which separates responsibilities into different layers for better organization:

1. Presentation Layer

This layer is responsible for displaying the user interface. It is built using HTML and CSS and includes:

- **Login page**
- **Message input area**
- **Countdown display**
- **Message reveal section**

The UI is designed using modern styles such as gradient backgrounds and animations to enhance user experience.

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2. Application Layer

This is the core logic layer implemented using JavaScript. It handles:

- **User authentication validation**
- **Countdown timer calculation**
- **Message locking and unlocking**
- **Event handling (button clicks, inputs)**

This layer ensures that the application behaves dynamically based on user actions.

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3. Storage Layer

This layer uses LocalStorage to store user data. It ensures:

- **Data persistence after page refresh**

- No need for database
- Fast access to stored data

4.3 Data Storage Design (LocalStorage)

The system uses LocalStorage to store data in key-value format. Each key represents a specific part of the application.

Data Structure:

- `loggedIn` → Stores whether the user is logged in
- `futureMessage` → Stores the message entered by the user
- `futureDate` → Stores the selected date and time

Advantages:

- No backend required
- Fast and efficient
- Data persists even after closing browser

4.4 System Requirements

4.4.1 Hardware Requirements

The system requires minimal hardware resources:

- Personal Computer or Laptop
- Minimum 2GB RAM
- Standard keyboard and mouse

Since the application is lightweight, it can run even on low-end systems.

4.4.2 Software Requirements

The software requirements include:

- Web Browser (Google Chrome, Edge, Firefox)
- Visual Studio Code (for development)

- **Live Server Extension (to run project)**

The application is compatible with all modern browsers.

🟦 5 SYSTEM IMPLEMENTATION (DETAILED)

5.1 System Methodology

The system is developed using a structured and iterative approach. Each stage of development focuses on a specific functionality.

Steps:

1. **Designing UI layout**
2. **Implementing login functionality**
3. **Adding message storage logic**
4. **Creating countdown timer**
5. **Testing and debugging**

This step-by-step approach ensures that errors are minimized and functionality is tested at each stage.

5.2 Module Descriptions

5.2.1 User Login and Authentication Module

This module ensures that only authorized users can access the application. It validates user credentials entered in the login form.

- **If credentials match → user is redirected**
- **If incorrect → error message shown**

The login status is stored in LocalStorage to maintain session.

5.2.2 Message Creation Module

This module allows users to write messages in a text area.

- **Users can type personal messages**
- **Messages are stored securely**
- **Prevents empty input submission**

This module forms the core functionality of the system.

5.2.3 Date Selection and Countdown Module

This module enables users to select a future date and time.

- **JavaScript calculates time difference**
- **Countdown is updated every second**
- **Displays days, hours, minutes, seconds**

This creates an interactive and real-time experience.

5.2.4 Message Unlock Module

The message remains hidden until the selected date is reached.

- **If time not reached → message locked**
- **If time reached → message displayed**

This feature adds uniqueness and emotional value to the application.

5.2.5 LocalStorage Data Management Module

This module handles storing and retrieving data.

Functions include:

- **Save data**
- **Retrieve data**
- **Update data**

It ensures that user data remains available even after page reload.

5.2.6 User Interface and Animation Module

This module improves visual appearance and user interaction.

Features:

- **Animated gradient background**
- **Button click effects**
- **Smooth transitions**
- **Responsive layout**

These features make the application modern and engaging.

🟦 6 RESULTS AND DISCUSSION (DETAILED)

6.1 System Output Screens

The application provides multiple screens:

- **Login Screen → User authentication**
- **Dashboard → Message input and date selection**
- **Countdown Display → Shows remaining time**
- **Message Reveal Screen → Displays message**

Each screen is designed to be user-friendly and visually appealing.

6.2 Performance Evaluation

The system performs efficiently because:

- **No server communication required**
- **Fast execution using JavaScript**
- **Instant UI updates**

The application shows minimal delay and works smoothly even on low-end devices.

6.3 System Testing

The system was tested under different conditions:

Test Cases:

- **Login validation**
- **Countdown accuracy**
- **Message storage**
- **Data persistence**

Results:

- **All tests passed successfully**
- **No major errors found**
- **System is stable and reliable**

🟦 7 CONCLUSION AND FUTURE WORK (DETAILED)

7.1 Conclusion

The *Future Me Website with Login System* successfully demonstrates the use of modern web technologies to create an interactive and meaningful application. The system allows users to securely store messages and retrieve them at a future time, providing both functional and emotional value.

The use of LocalStorage eliminates the need for a backend database, making the system simple, efficient, and easy to deploy. The integration of login authentication ensures basic security and controlled access.

Overall, the project achieves its objectives and serves as a strong example of frontend web development.

7.2 Future Enhancements

Although the system works effectively, it can be further improved with advanced features:

- **Add Signup system for multiple users**
- **Integrate database (Firebase) for real storage**
- **Add email or notification alerts**
- **Improve security using encryption**
- **Make application fully mobile responsive**
- **Add cloud storage support**

These enhancements will make the system more scalable and suitable for real-world applications

REFERENCES

[1] Mozilla Developer Network (MDN), “JavaScript Documentation,” Available: https://developer.mozilla.org

[2] W3C, “HTML Standard,” World Wide Web Consortium, Available: https://www.w3.org

[3] W3Schools, “HTML, CSS, and JavaScript Tutorials,” Available: https://www.w3schools.com

[4] Google Developers, “Web APIs Documentation,” Available: https://developers.google.com

[5] Flanagan, D., *JavaScript: The Definitive Guide*, 7th Edition, O’Reilly Media, 2020

[6] Nielsen, J., *Usability Engineering*, Morgan Kaufmann, 1993

[7] WHATWG, “HTML Living Standard,” Available: https://html.spec.whatwg.org

[8] ECMA International, “ECMAScript Language Specification,” Available: https://tc39.es/ecma262

[9] MDN Web Docs, “Web Storage API (LocalStorage),” Available: https://developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API

[10] Google Chrome Developers, “Web Storage and Client-Side Data,” Available: https://developer.chrome.com/docs

APPENDIX–1: SOURCE CODE PSEUDOCODE

The *Future Me Website with Login System* is a web-based application designed to allow users to store messages for their future selves and retrieve them at a specified date and time. The system includes login authentication, countdown tracking, and message unlocking functionality. It is built using HTML, CSS, and JavaScript, and uses LocalStorage for data persistence.

🔹 System Initialization and Message Handling Logic

START

1. **Load `login.html` or `index.html` in the browser.**
2. **Initialize System Variables:**
   - `message = ""` **(Load from LocalStorage)**
   - `targetDate = null` **(Load from LocalStorage)**
   - `isLoggedIn = false`
3. **Check Authentication State:**
   - **IF User is not logged in THEN**
     → Display Login Page.
   - **ELSE**
     → Display Future Me Dashboard.
4. **Login Process:**
   - **IF User enters username and password**
     → Validate credentials
     → IF correct THEN
     - **Set `isLoggedIn = true`**
     - **Save login state in LocalStorage**
     - **Redirect to Dashboard**
       → ELSE
     - **Display error message**
5. **Display Message Dashboard:**
   - **Show text area for message input**
   - **Show date/time picker**
   - **Show countdown section**
   - **Show locked message placeholder**
6. **Save Message:**

   - IF User clicks "Save" THEN
     → Capture message and selected date
     → Store in LocalStorage
     → Initialize countdown
7. Countdown Logic:
   - EVERY 1 second:
     → Get current system time
     → Calculate difference between current time and targetDate
     → Display remaining time
8. Message Unlock Logic:
   - IF current time >= targetDate THEN
     → Display saved message
     → Stop countdown timer
9. Additional Actions:
   - IF User clicks "Preview" THEN
     → Show saved message in alert
   - IF User clicks "Reset" THEN
     → Clear LocalStorage
     → Reload page
   - IF User clicks "Logout" THEN
     → Remove login status
     → Redirect to login page

END

📑 APPENDIX–2: TEST CASES AND SYSTEM SUMMARY

The *Future Me Website with Login System* was tested under various scenarios to ensure correct functionality of login, message storage, countdown, and message unlocking.

🔹 Test Case 1: Login Validation

- Description: User enters valid credentials.
- Expected Result: User is redirected to dashboard.
- Actual Result: Login successful and dashboard loaded.
- Status: PASS ✅

🔹 Test Case 2: Message Storage

- **Description: User enters a message and selects a future date.**
- **Expected Result: Message and date are stored in LocalStorage.**
- **Actual Result: Data stored successfully and persists after refresh.**
- **Status: PASS ✅**

🔹 Test Case 3: Countdown Functionality

- **Description: System calculates time remaining.**
- **Expected Result: Countdown updates every second accurately.**
- **Actual Result: Countdown displayed correctly with no delay.**
- **Status: PASS ✅**

🔹 Test Case 4: Message Unlock

- **Description: Current time reaches selected date.**
- **Expected Result: Message becomes visible automatically.**
- **Actual Result: Message displayed correctly after countdown ends.**
- **Status: PASS ✅**

🔹 Test Case 5: Data Persistence

- **Description: User refreshes or reopens browser.**
- **Expected Result: Saved message and date remain intact.**
- **Actual Result: LocalStorage successfully restores data.**
- **Status: PASS ✅**

📊 Final System Summary

- **Total Test Cases Executed: 5**

- **Passed: 5**
- **Failed: 0**
- **System Accuracy: High reliability in countdown timing and data storage**

🔹 Final System Status

The *Future Me Website with Login System* has been successfully developed and tested. The application effectively combines login authentication, message storage, and time-based unlocking functionality. It provides a simple, interactive, and meaningful user experience.